User Guide

Project: SongDB Music Library

Author: Mohamed Swesa

Neptun Code: D044RQ

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1. Goal of the Program

The **SongDB Music Library Manager** allows you to organize your music collection digitally. It serves as a personal database where you can:

- **Store** details about your favorite songs (Artist, Album, Year, Genre, etc.).
- **Add** new songs to your database by providing data for each field.
- **View** your collection sorted by different criteria.
- **Filter** songs to find specific artists or albums.
- **Import** lists of songs from other files to merge collections.
- **Delete** song from your database by providing the name of the song.
- **Save** your changes permanently to a file.

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2. Instructions: How to Use the Program

When you run the program, the "Main Menu" will appear. Type the number corresponding to the action you want to perform and press **Enter**.

1. Add New Song

- Select option **1**.
- Type the requested details (Title, Performer, Album, Year, Genre, Length).
- **Note:** Start every word with a capital letter.
- **Note:** Year and Length must be valid positive numbers.

2. Display All Songs (Sorted)

- Select option **2**.
- A sub-menu will ask how you want to sort the view:
    - **1:** Alphabetical by Title.

    - **2:** Alphabetical by Performer.
    - **3:** Release Year (Newest to Oldest).
- The program will sort your library and print the entire list.

3 - 6. List/Search Options

- These options act as **filters**.
- Select option **3** (Artist), **4** (Album), **5** (Year), or **6** (Genre).
- Enter the keyword (e.g., "Rock" for genre or "Queen" for artist).
- The program will list *only* the songs that match your search.

7. Delete Song

- Select option **7**.
- Enter the **Title** of the song you wish to remove.
- If multiple songs have the same title, you will be shown a list and asked to select the specific number to delete.
- You will be asked to confirm (y/n) before the song is permanently removed.

8. Import Songs from File (Merge)

- Select option **8**.
- Enter the filename of another song list (e.g., extra_songs.txt).
- The program will read the songs from that file and **add them** to your current library.
- **Tip:** This is useful for combining two different playlists.

9. Save and Quit

- Select option **9** when you are finished.
- The program will save **all** your songs (including new additions and imports) to the main database file.
- It will then close the program safely.

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3. Where Files are Saved and Loaded

The Default Database: songs.txt

- **Loading:** When the program starts, it automatically looks for a file named songs.txt in the same folder as the program executable. If found, it loads all songs into memory.

- **Saving:** When you choose "Save and Quit" (Option 9), the program overwrites songs.txt with your current list.

Import Files

- You can create your own text files (e.g., my_playlist.txt) to import using **Option 8**.
- These files must follow the strict **7-line format** for each song:
    1. Title
    2. Performer
    3. Album
    4. Year
    5. Genre
    6. Minutes
    7. Seconds